

SHUBHAM LOKARE

Generative AI Engineer | AI/ML Engineer | LLM & NLP Specialist

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Profile Summary

Microsoft Certified AI Engineer with 1.8+ years of professional experience and 3+ years of hands-on expertise, all focused on Generative AI, Large Language Models, and Agentic Systems. Architected production-grade RAG pipelines, multi-agent systems, and MLOps workflows driving a 28% improvement in query accuracy and 35% increase in user adoption. Experienced across LangChain, LangGraph, MCP, OpenAI, and Hugging Face with strong foundations in LLM fine-tuning (LoRA, QLoRA). Proficient in vector databases (Pinecone, Chroma) and cloud deployment on Azure and AWS.

Technical Skills

- **Generative AI / AI:** Chatbots, LLMs, RLHF, RAG Pipeline, Agentic AI, NLP, Transformers, Embeddings, Multimodal AI, Vector Databases (Pinecone, FAISS, Weaviate).
- **Prompt Engineering & Fine-Tuning:** LoRA, QLoRA, Prompt Engineering, Few-Shot Learning, ReAct, Chain-of-Thought (CoT), Tree-of-Thought (ToT), RAG Evaluation (RAGAS).
- **ML/DL:** Regression, Random Forest, XGBoost, LightGBM, LSTM, Computer Vision, YOLO, Clustering, Dimensionality Reduction, Ensemble Learning.
- **Frameworks & Tools:** LangChain, LangGraph, LangSmith, AutoGen, Hugging Face, MCP, LlamaIndex, Haystack, OpenAI API, Ollama API, Cohere API, Claude API.
- **Backend & Libraries:** Fast API, REST APIs, WebSocket, Async APIs, API Integration, Microservices Architecture, Python, SQL, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy.
- **Platforms & DevOps:** AWS, Azure, Azure OpenAI, Azure AI Foundry, Azure ML, Docker, Git, Nginx.

Professional Experience

AI Engineer

ImmersiveVision Technology Pvt.Ltd | Dec 2024 – Present

- **Led end-to-end AI development** for Cadaviz – a 3D Virtual Human Dissection Table, including design, development, and deployment of RAG pipelines, chatbots (OpenAI, Claude API), MCP workflows, and Whisper-based voice solutions improving query response accuracy by 28%.
- Built and integrated chatbots for feedback and concept explanation, enabling medical students to interact with the dissection table through Q&A, search, and voice commands — increasing user adoption by 35%.
- Developed and fine-tuned Generative AI, NLP, and computer vision models for object detection, radiology/anatomy analysis, multimodal processing — boosting diagnostic accuracy by 18%.
- Conducted data analysis, model optimization, and accuracy improvements, delivering real-time dashboards (Power BI) — cutting report generation time by 60%.
- Managed full-stack AI deployment using AWS, Docker, and FastAPI, while reducing deployment cycle time by 70% and mentoring 4+ engineers in AI workflows.

Data Scientist Intern

AISPRA | July 2024 – Dec 2024

- Conducted comprehensive EDA, data preprocessing, and feature engineering, optimizing datasets to improve ML/DL model performance by 10–15%.
- Developed and fine-tuned ML models (Random Forest, XGBoost, LightGBM, SVM) and DL models (ANN, CNN, RNN, LSTM, YOLO), achieving high accuracy across classification and forecasting tasks.

Selected Projects

DataSynapse AI – Multi-Agent AI Platform with RLHF, RAG & MCP

[GitHub](#)

Problem Solved: Fragmented AI workflows force engineers to spend **60–70%** of their time manually coordinating tasks **across ML/DL, NLP, code execution, and visualization, causing inefficiency and delays.**

- Designed and implemented a **production-ready 10-agent AI system** with Core Orchestrator & MCP Dispatcher, automating workflows across **EDA, data processing, ML/DL, NLP and visualization.**
- Leveraged **LangGraph** for multi-agent coordination, achieving **~90% automation**, optimizing **routing, task execution, and parallel processing** for AI engineers and data scientists reducing manual engineering effort by **65%.**
- Integrated **RLHF (Reinforcement Learning from Human Feedback)** to continuously improve agent performance and workflow efficiency.
- Built **RAG pipelines** using external knowledge sources (**Wikipedia, APIs, web search**) with **vector memory** for **context-aware, real-time responses.**
- Applied **advanced prompting techniques (ReAct, CoT, ToT)** and implemented **self-attention mechanisms** to improve query understanding, reasoning by **22%.**
- Developed **FastAPI web interface** and CLI runner, enabling scalable, high-performance AI system engineering with memory management and robust logging — cutting experiment cycle time by **40%**
- Applied **LLM fine-tuning (LoRA, QLoRA)** and **enterprise-grade system** optimizations — enabling scalable, **high-performance multi-agent orchestration across complex AI workflows.**

Medical ChatBot – AI-Powered Clinical Assistant with RAG & NLP

[GitHub](#)

Problem Solved: Manual review of clinical reports is **slow** and error-prone; medical professionals need AI-assisted tools **for faster diagnosis, treatment planning, and reporting.**

- Built an **AI-powered medical chatbot** to process clinical reports (PDF, DOCX, JSON, CSV) — reducing manual review time by **50%.**
- Developed a **RAG pipeline** using FAISS vector database for context-aware, real-time responses from medical data, improving response accuracy by **23%.**
- Designed a **Flask + React web interface** with interactive chat, file upload, and automated PDF report generation — enabling clinicians to generate reports in **under 2 minutes.**
- Integrated advanced **NLP models** (Hugging Face Transformers, Cohere, LangChain) for **intelligent diagnosis and medical recommendations** — increasing **recommendation relevance by 20%.**
- Containerized the chatbot with **Docker, enabling scalable real-time querying** (e.g., clinicians upload reports and immediately ask **“What is the diagnosis for this patient?”**).

Certifications / Training

- Microsoft Azure AI Engineer Associate [Link](#)
- Professional Program on Data Science and AI (360DigiTMG) [Link](#)
- Microsoft Power BI [Link](#)

Education

- **B.E in Computer Science Engineering | 2018 – 2022**
Savitribai Phule Pune University, Pune
- **12th Standard (HSC) | 2017 – 2018**
Shree Venkatesh Mahavidyalaya, Walandi